

Apollo 14: Mission Details



NASA

Jul 08, 2009

Article

Contents

- Mission Objective
- Mission Highlights
- Crew
- Backup Crew
- Payload
- Prelaunch Milestones
- Launch
- Orbit
- Landing

Mission Objective

The primary objectives of this mission were to explore the Fra Mauro region centered around deployment of the Apollo Lunar Surface Scientific Experiments Package, or ALSEP; lunar field geology investigations; collection of surface material samples for return to Earth; deployment of other scientific instruments not part of ALSEP; orbital science involving high-resolution photography of candidate future landing sites; photography of deep-space phenomena, such as zodiacal light and gegenschein; communications tests using S-band and VHF signals to determine reflective properties of the lunar surface; engineering and operational evaluation of hardware and techniques; tests to determine variations in S-band signals; and photography of surface details from 60 nautical miles in altitude.



This is the Apollo 14 crew patch designed by astronauts Alan B. Shepard Jr., commander; Stuart A. Roosa, command module pilot; and Edgar D. Mitchell, lunar module pilot. It features the astronaut lapel pin approaching the moon and leaving a comet trail from the liftoff point on Earth.

NASA

Mission Highlights

Apollo 14 launched at 4:03 p.m. EST. At approximately 3:41 p.m. ground elapsed time, or GET, difficulties were experienced in docking with the lunar module, or LM, and six attempts were required before a “hard dock” was achieved. An in-flight televised inspection of the docking mechanism revealed no apparent reason for the malfunction and the system appeared to be functioning normally.

Prior to the powered descent initiation, or PDI, for the Antares landing, a short in the LM computer abort switch was discovered, which could have triggered an undesired abort

during the LM's descent. On Feb. 5, Antares made the most precise landing to date, approximately 87 feet from the targeted landing point. The landing point coordinates were 3 degrees, 40 minutes, 27 seconds south and 17 degrees, 27 minutes, 58 seconds west, midway between the Doublet and Triplet craters in the hilly uplands of the Fra Mauro crater, and about 110 miles east of the Apollo 12 landing site.

Due to communications system problems, the first period of extra vehicular activity, or EVA, began almost one hour later than scheduled with Commander Alan Shepard setting foot on the lunar surface at 114 hours, 31 minutes GET. The first of the two EVA periods included ALSEP deployment and lasted four hours, 49 minutes.

The second EVA on Feb. 6 began when the LM egress hatch was opened at 4:15 a.m. EST. During this EVA, Shepard and Edgar Mitchell moved more than half a mile from their LM, conducting selenological investigations, collecting samples and attempting to reach the rim of Cone crater, approximately 300 feet above the landing site. NASA personnel monitoring the EVA estimate that the two astronauts were within 50 to 75 meters of the crater rim when they were advised by mission control to collect samples at that spot and begin their traverse back to the LM. The second EVA lasted four hours, 35 minutes, resulting in a new mark for EVA time by a lunar landing crew: nine hours, 24 minutes. Shepard set a new distance-traveled record on the lunar surface of approximately 9,000 feet.

During the two traverses, the astronauts collected 94 pounds of rocks and soil for return to Earth. The samples were scheduled to go to 187 scientific teams in the United States, as well as 14 other countries for study and analysis.

Orbital science activities were conducted by Stuart Roosa during the lunar surface activities period. He experienced some difficulties with the high-resolution, motion-compensating Hycon Lunar Topographic Camera while attempting to photograph the Descartes area, the landing site planned for Apollo 16.

The liftoff of Antares from the lunar surface took place precisely on schedule. Rendezvous and docking occurred only two minutes later than scheduled. The command module Kitty Hawk splashed down safely in the Pacific Ocean at 4:05 p.m. EST Feb. 9, exactly nine days and two minutes after launch. The actual landing point was only 1.02 nautical miles off its targeted point of about 765 nautical miles south of Samoa, and four miles from the prime recovery ship, the USS New Orleans. The mission duration from liftoff to splashdown was 216 hours, two minutes.

Crew

Alan B. Shepard Jr., Commander

Edgar D. Mitchell, Lunar Module Pilot

Stuart A. Roosa, Command Module Pilot

Backup Crew

Eugene A. Cernan, Commander

Joe H. Engle, Lunar Module Pilot

Ronald E. Evans, Command Module Pilot

Payload

Kitty Hawk (CM-110)

Antares (LM-8)

Prelaunch Milestones

1/21/70 – S-IVB ondock at Kennedy

1/21/70 – S-II ondock at Kennedy

1/12/70 – S-IC ondock at Kennedy

5/6/70 – S-IU ondock at Kennedy

Launch

Jan. 31, 1971; 4:03 p.m. EDT

Launch Pad 39A

Saturn-V AS-509

High Bay 3

Mobile Launcher Platform-2

Firing Room 2

Orbit

Altitude: 118.55 miles

Inclination: 31.12 degrees

Orbits: 34 revolutions

Duration: nine days, two minutes

Distance: 1,150,321 miles

Lunar Location: Fra Mauro

Lunar Coordinates: 3.65 degrees south, 17.48 degrees west

Landing

Feb. 9, 1971

Pacific Ocean

Recovery Ship: USS New Orleans